

module4lab2

September 27, 2026

```
[1]: import pandas as pd
```

```
[2]: df=pd.read_csv("Downloads/allbacks.csv")
```

```
[3]: df.head()
```

```
[3]:
```

	rownames	volume	area	weight	cover
0	1	885	382	800	hb
1	2	1016	468	950	hb
2	3	1125	387	1050	hb
3	4	239	371	350	hb
4	5	701	371	750	hb

```
[4]: df.shape
```

```
[4]: (15, 5)
```

```
[5]: df.info()
```

```
<class 'pandas.DataFrame'>  
RangeIndex: 15 entries, 0 to 14  
Data columns (total 5 columns):  
#   Column      Non-Null Count  Dtype  
---  -  
0   rownames    15 non-null     int64  
1   volume      15 non-null     int64  
2   area        15 non-null     int64  
3   weight      15 non-null     int64  
4   cover       15 non-null     str  
dtypes: int64(4), str(1)  
memory usage: 762.0 bytes
```

```
[6]: df.info
```

```
[6]: <bound method DataFrame.info of
```

	rownames	volume	area	weight	cover
0	1	885	382	800	hb
1	2	1016	468	950	hb
2	3	1125	387	1050	hb
3	4	239	371	350	hb

4	5	701	371	750	hb
5	6	641	367	600	hb
6	7	1228	396	1075	hb
7	8	412	0	250	pb
8	9	953	0	700	pb
9	10	929	0	650	pb
10	11	1492	0	975	pb
11	12	419	0	350	pb
12	13	1010	0	950	pb
13	14	595	0	425	pb
14	15	1034	0	725	pb>

```
[7]: df.describe()
```

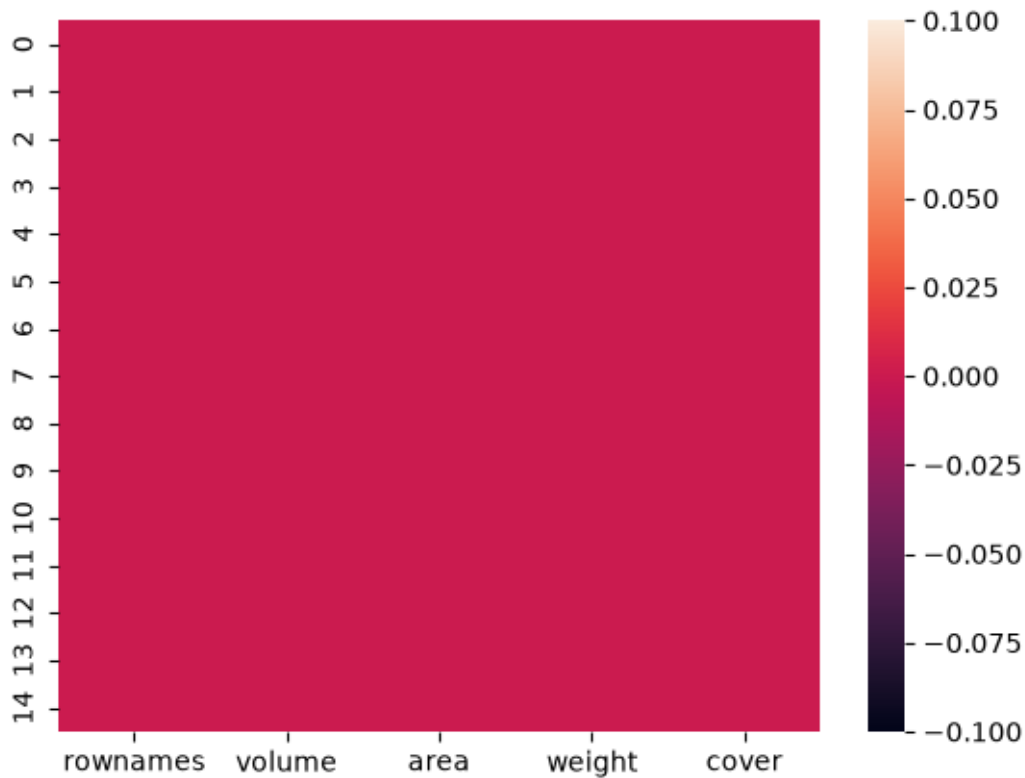
```
[7]:
```

	rownames	volume	area	weight
count	15.000000	15.000000	15.000000	15.000000
mean	8.000000	845.266667	182.800000	706.666667
std	4.472136	339.879909	203.586837	268.837887
min	1.000000	239.000000	0.000000	250.000000
25%	4.500000	618.000000	0.000000	512.500000
50%	8.000000	929.000000	0.000000	725.000000
75%	11.500000	1025.000000	376.500000	950.000000
max	15.000000	1492.000000	468.000000	1075.000000

```
[8]: import seaborn as sns
```

```
[9]: sns.heatmap(df.isnull())
```

```
[9]: <Axes: >
```



```
[10]: #median cover type
```

```
[11]: print(df['cover'].value_counts())
```

```
cover
pb      8
hb      7
Name: count, dtype: int64
```

```
[12]: #median book will be pb as there are 15 book spots there will be equal amounts
      ↳ of books of hb and pb with the remainder pb being the median
```

```
[13]: mean_all = df['weight'].mean()
      print(mean_all)
```

```
706.6666666666666
```

```
[14]: #mean weight of all books match with the df.describe for total books
```

```
[15]: mean_pb=df[df['cover']=='pb']['weight'].mean()
      print(mean_pb)
```

```
628.125
```

```
[16]: #paperbacks weight are less than the average
```

```
[17]: mean_hb=df[df['cover']=='hb']['weight'].mean()  
      print(mean_hb)
```

```
796.4285714285714
```

```
[18]: #hard books weight are more than the average
```

```
[19]: #looking at the group means,a hard cover book weighs more than the paperbacks
```